



CLINICAL PROGRAMMING BOOTCAMP • MODULE 14

# From Tabulation to Analysis

Why ADaM exists — and building ADSL

You have spent ten days making data faithful. Now you make it answerable.

*Hands-on: Notebook 14 (SAS) — Build ADSL*

# By the end of this module you can...

1

**Say why ADaM exists** in one sentence, without using the word 'analysis-ready' as if it explained itself.

2

**Name the three dataset classes** ADSL, BDS and OCCDS — and say which to reach for.

3

**Build ADSL end to end** treatment variables, population flags, disposition, baseline measurements.

4

**Apply the ADaM baseline rule** and say why copying VSBLFL breaks silently.

5

**Defend traceability** explain how a regulator gets from a number in a table back to a CRF page.

# The same subject, told two different ways

Subject ABC-01-01-004 wants to appear in a table of mean treatment duration by arm.

## SDTM asks: what happened?

**DM** → RFXSTDTC, RFXENDTC (text)

**DS** → disposition, why they left

**EX** → what was actually given

*Three domains. Nothing derived.*

*To get the duration you must join  
them and compute it yourself —  
every time, in every program.*

## ADaM asks: what do I report?

**ADSL** → one row, everything on it

TRT01P = Placebo

TRTSDT = 06MAR2024

TRTEDT = 25MAR2024

TRTDURD = 20

SAFFL = Y

*One dataset. PROC MEANS. Done.*

*Same subject. Same truth. Different organising principle — because they have different readers.*

## THE TEST A DATASET MUST PASS

**A table, listing or figure can be produced by subsetting and summarising ONE dataset — with no further merging and no further derivation.**

If you need a join at the analysis step, the analysis dataset was built wrong.

That is not a style preference. It is the difference between a result a reviewer can reproduce and one they cannot.

# Three dataset classes — and how to choose

## ADSL

Subject-Level

**ONE row per subject**

The spine. Treatment, populations, key dates.  
Every other dataset copies from it.

## BDS

Basic Data Structure

**One row per subject  
per parameter per visit**

Anything measured repeatedly:  
vitals, labs, scores. ADVS, ADLB.

## OCCDS

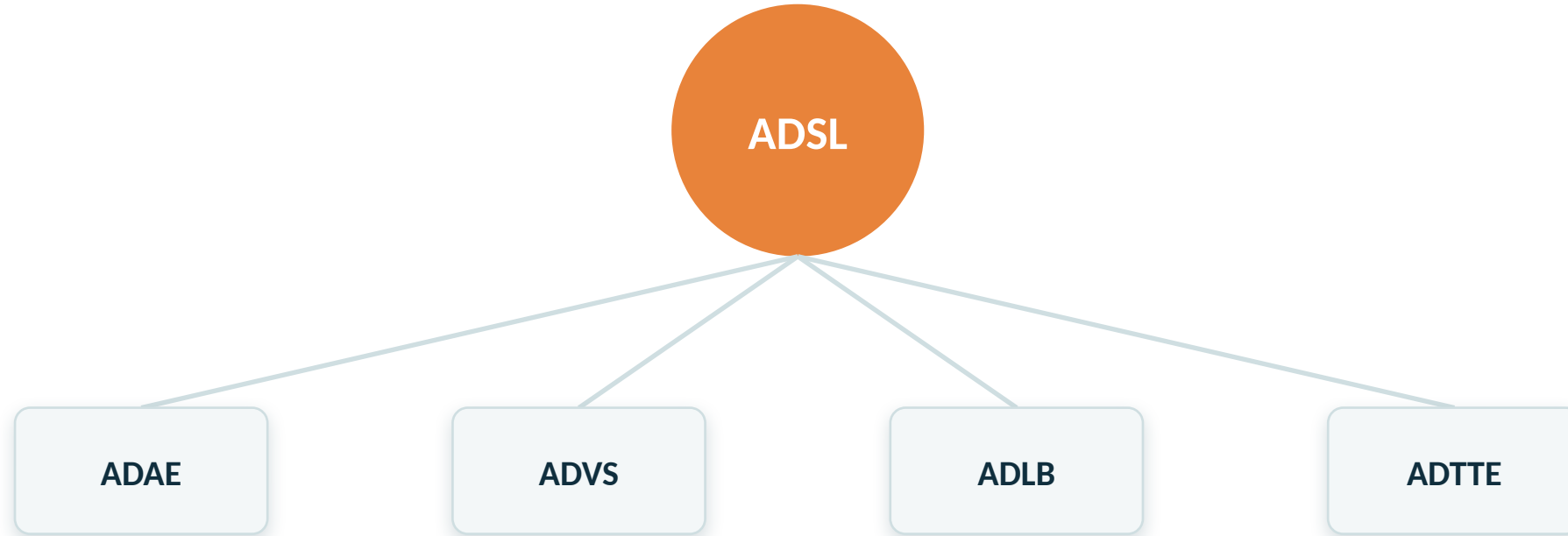
Occurrence

**One row per event**

Things that either happened or did not:  
adverse events, medications. ADAE.

*Pick by SHAPE, not by subject matter. A questionnaire score is BDS. A protocol deviation is OCCDS.*

# One dataset defines the study's subject-level truth



TRT01P · TRT01A · SAFFL · ITTFL · TRTSDT · TRTEDT · AGEGR1

*These are COPIED downstream — never re-derived. Re-deriving invites the two to drift, and nothing tells you when they do.*

# In ADaM, a date is a number

## SDTM

```
RFXSTDTC = "2024-03-01"
```

```
CHARACTER. ISO 8601.
```

```
Faithful, readable, sortable.
```

```
You cannot subtract it.
```

## ADaM

```
TRTSDT = 23436
```

```
(01MAR2024)
```

```
NUMERIC SAS date.
```

```
TRTEDT - TRTSDT + 1 = 28
```

**INPUT ()** converts character → numeric. **PUT ()** converts numeric → character.

*Getting these backwards is the most common SAS error in this course. It does not always fail loudly.*

*Why the change? Because a table needs arithmetic, and arithmetic needs numbers.*

# Baseline is a question about time, not about a visit

**Baseline = the LAST non-missing value on or before the date of first dose.**

## The tempting shortcut

```
if vsblfl = 'Y';
```

HEIGHT is collected only at SCREENING.  
VSBLFL marks the BASELINE visit.

Zero height records carry VSBLFL='Y'.  
All 8 subjects lose baseline height —  
and their BMI. No error. No warning.

## The rule, done properly

```
where vsdte <= trtsdt  
... if last.vstestcd;
```

HEIGHT → resolves to Screening  
WEIGHT → resolves to Baseline

Same rule. Different visits.  
Because it asks about TIME.

*A rule written in terms of visit labels breaks the moment a parameter is collected on a different schedule.*

# Different questions that happen to have the same answer

|                                 |                                                                      |             |
|---------------------------------|----------------------------------------------------------------------|-------------|
| <b>SAFFL</b><br>Safety          | Did the subject take at least one dose?<br><br>TRTSDT is not missing | Y for all 8 |
| <b>ITTFL</b><br>Intent-to-treat | Was the subject randomised?<br><br>RANDDT is not missing             | Y for all 8 |
| <b>COMPLFL</b><br>Completers    | Did the subject finish the study?<br><br>DS disposition = COMPLETED  | Y for all 8 |

All three are 'Y' for every ABC-01 subject — and all three are still derived from different rules.

*A screen failure, or a subject randomised but never dosed, separates them. Then the tables built on each separate too.*

# A regulator must get from a table back to a CRF page



Every link must be documented, and every derivation must be reproducible from the link below it.

**This is why ADaM carries SRCDOM / SRCVAR / SRCSEQ.** A record that names its source SDTM row can be checked. A derived record with no single source leaves them blank — and that blank is information, not an omission.

# Six ADSL mistakes, and what each one costs

**Copying VSBLFL as baseline**

Every subject loses baseline height and BMI. No error.

**Re-deriving SAFFL downstream**

Two answers to 'who was dosed'. Neither defensible.

**DCSREAS filled for completers**

'Discontinued for reason X' becomes uncountable.

**TRTDURD without the +1**

Every duration off by one; a one-day subject reads as zero.

**Duplicate USUBJID**

Silently multiplies rows in every downstream merge.

**Dates left as character**

Arithmetic impossible; study day cannot be derived.

*Five of these six produce no error message. That is the recurring lesson of this course.*

## WHAT'S NEXT

# Build the spine first

1

**Step 1 • Notebook 14** Build ADSL from DM, DS and VS. Eight rows, and one of them discontinued.

2

**Step 2 • Exercise 1** Do the baseline the WRONG way on purpose. Watch it fail silently.

3

**Step 3 • Check** PROC COMPARE against data/adam/adsl.csv — the finished reference.

*Everything in the next four days copies from what you build today. Get ADSL right and the rest is bookkeeping.*